

$$H = \sum_k d_k^\dagger H_k d_k \quad (1)$$

$$d_k^\dagger = \begin{pmatrix} d_{k,x}^\dagger & d_{k,y}^\dagger & d_{k,z}^\dagger \end{pmatrix}, \quad d_k = \begin{pmatrix} d_{k,x} \\ d_{k,y} \\ d_{k,z} \end{pmatrix}, \quad H_k = \dots \quad (2)$$

$$d_{\mathbf{r},\mu} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} d_{\mathbf{k},\mu}, \quad d_{\mathbf{r},\mu}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} d_{\mathbf{k},\mu}^\dagger \quad (3)$$

$$d_{\mathbf{k},\mu} = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k}\cdot\mathbf{r}} d_{\mathbf{r},\mu}, \quad d_{\mathbf{k},\mu}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{i\mathbf{k}\cdot\mathbf{r}} d_{\mathbf{r},\mu}^\dagger \quad (4)$$

$$\begin{aligned} H_{\mathbf{k}} &= U_{\mathbf{k}} D_{\mathbf{k}} U_{\mathbf{k}}^\dagger \\ &= \begin{pmatrix} \phi_{\mathbf{k},1} & \dots & \phi_{\mathbf{k},3N} \end{pmatrix} \text{diag} (E_{\mathbf{k},1}, \dots, E_{\mathbf{k},3N}) \begin{pmatrix} \phi_{\mathbf{k},1}^\dagger \\ \vdots \\ \phi_{\mathbf{k},3N}^\dagger \end{pmatrix} \end{aligned} \quad (5)$$

$$\begin{aligned} H &= \sum_k d_k^\dagger H_k d_k \\ &= \sum_k d_k^\dagger U_k D_k U_k^\dagger d_k \\ &= \sum_k \beta_k^\dagger D_k \beta_k \end{aligned} \quad (6)$$

$$\beta_k^\dagger \equiv d_k^\dagger U_k \quad (7)$$

$$\begin{aligned} \beta_{k,i}^\dagger &= d_{k,i}^\dagger \phi_{k,i} \\ &= \begin{pmatrix} d_{k,x}^\dagger & d_{k,y}^\dagger & d_{k,z}^\dagger \end{pmatrix} \begin{pmatrix} \phi_{k,i,x} \\ \phi_{k,i,y} \\ \phi_{k,i,z} \end{pmatrix} \\ &= \sum_{\mu \in \{x,y,z\}} d_{k,\mu}^\dagger \phi_{k,i,\mu} \\ &= \sum_{\mu} \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{i\mathbf{k}\cdot\mathbf{r}} d_{\mathbf{r},\mu}^\dagger \phi_{k,i,\mu} \\ &= \frac{1}{\sqrt{N}} \sum_{\mathbf{r},\mu} e^{i\mathbf{k}\cdot\mathbf{r}} d_{\mathbf{r},\mu}^\dagger \phi_{\mathbf{k},i,\mu} \end{aligned} \quad (8)$$

$$|\Psi\rangle \sim |\Omega\rangle + c_{\mathbf{k}} \beta_{\mathbf{k},1}^\dagger |\Omega\rangle \quad (9)$$

$$\langle \Psi | \sigma_{\mathbf{r}}^\mu | \Psi \rangle \sim c_{\mathbf{k}} \left\langle \Omega \left| \sigma_{\mathbf{r}}^\mu \beta_{\mathbf{k},1}^\dagger \right| \Omega \right\rangle + \text{c.c.} \quad (10)$$

$$\mathbf{r} \in A$$

$$\begin{aligned}
\langle \Omega | \sigma_{\mathbf{r}}^{\mu} \beta_{\mathbf{k},1}^{\dagger} | \Omega \rangle &= \left\langle \Omega \left| \sigma_{\mathbf{r}}^{\mu} \left(\frac{1}{\sqrt{N}} \sum_{\mathbf{r}',\nu} e^{i\mathbf{k}\cdot\mathbf{r}'} d_{\mathbf{r}',\nu}^{\dagger} \phi_{\mathbf{k},1,\nu} \right) \right| \Omega \right\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}',\nu} e^{i\mathbf{k}\cdot\mathbf{r}'} \langle \Omega | \sigma_{\mathbf{r}}^{\mu} d_{\mathbf{r}',\nu}^{\dagger} | \Omega \rangle \phi_{\mathbf{k},1,\nu} \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}',\nu} e^{i\mathbf{k}\cdot\mathbf{r}'} \langle 0_{\chi}, 0_{\alpha(0)} | (i b_{\mathbf{r}}^{\mu} c_{\mathbf{r}}) \alpha^{\dagger}(\mathbf{r}', \nu) \chi_{\mathbf{r}',\nu}^{\dagger} | 0_{\chi}, 0_{\alpha(\mathbf{r}',\nu)} \rangle \phi_{\mathbf{k},1,\nu} \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}',\nu} e^{i\mathbf{k}\cdot\mathbf{r}'} \langle 0_{\chi}, 0_{\alpha(0)} | i (\chi_{\mathbf{r},\mu} + \chi_{\mathbf{r},\mu}^{\dagger}) c_{\mathbf{r}} \alpha^{\dagger}(\mathbf{r}', \nu) \chi_{\mathbf{r}',\nu}^{\dagger} | 0_{\chi}, 0_{\alpha(\mathbf{r}',\nu)} \rangle \phi_{\mathbf{k},1,\nu} \quad (11) \\
&= \frac{1}{\sqrt{N}} e^{i\mathbf{k}\cdot\mathbf{r}} \langle 0_{\chi}, 0_{\alpha(0)} | i \chi_{\mathbf{r},\mu} c_{\mathbf{r}} \alpha^{\dagger}(\mathbf{r}, \mu) \chi_{\mathbf{r},\mu}^{\dagger} | 0_{\chi}, 0_{\alpha(\mathbf{r},\mu)} \rangle \phi_{\mathbf{k},1,\mu} \\
&= \frac{1}{\sqrt{N}} e^{i\mathbf{k}\cdot\mathbf{r}} \langle 0_{\chi}, 0_{\alpha(0)} | i c_{\mathbf{r}} \alpha^{\dagger}(\mathbf{r}, \mu) | 0_{\chi}, 0_{\alpha(\mathbf{r},\mu)} \rangle \phi_{\mathbf{k},1,\mu} \\
&\equiv \frac{1}{\sqrt{N}} e^{i\mathbf{k}\cdot\mathbf{r}} \langle \Psi(0) | i c_{i,A} \alpha^{\dagger}(2) | \Psi(2) \rangle \phi_{\mathbf{k},1,\mu} \\
&= \frac{1}{\sqrt{N}} e^{i\mathbf{k}\cdot\mathbf{r}} \langle \Psi(0) | \Psi(2) \rangle \langle i c_{i,A} \alpha^{\dagger}(2) \rangle \phi_{\mathbf{k},1,\mu}
\end{aligned}$$

$$\bullet \langle \Psi(1) | \Psi(2) \rangle$$

$$\langle \Psi(1) | \Psi(2) \rangle = \tilde{N}(1) \tilde{N}(2) (-1)^{N(N+1)/2} \text{Pf}(M) \quad (12)$$

$$M \equiv \begin{pmatrix} \tilde{F}(2) & -I \\ I & -\tilde{F}^*(1) \end{pmatrix} \quad (13)$$

$$\tilde{N} = \left| \det \left(\widetilde{W} \right) \right|^{1/2} \quad (14)$$

$$\tilde{F} = \tilde{V}^* \left(\widetilde{W}^* \right)^{-1} \quad (15)$$

$$\tilde{U} \equiv U^{\dagger}(0)U, \quad \tilde{U}^{\dagger} = U^{\dagger}U(0) \quad (16)$$

$$\tilde{U} = \begin{pmatrix} \widetilde{W} & \tilde{V}^* \\ \tilde{V} & \widetilde{W}^* \end{pmatrix}, \quad \tilde{U}^{\dagger} = \begin{pmatrix} \widetilde{W}^{\dagger} & \tilde{V}^{\dagger} \\ \tilde{V}^{\top} & \widetilde{W}^{\top} \end{pmatrix} \quad (17)$$

$$\tilde{U}(0) = I, \quad \tilde{U}^{\dagger}(0) = I \quad (18)$$

$$\widetilde{W}(0) = I, \quad \tilde{V}(0) = 0 \quad (19)$$

$$\tilde{N}(0) = 1/\sqrt{2} \quad (20)$$

$$\tilde{F}(0) = 0 \quad (21)$$

$$M = \begin{pmatrix} \tilde{F}(2) & -I \\ I & 0 \end{pmatrix} \quad (22)$$

- $\circ \langle c_{i,A} \alpha^\dagger(2) \rangle$

$$\boxed{\alpha^\dagger(2) = \alpha^\dagger(0) \widetilde{W}(2) + \alpha^\top(0) \widetilde{V}(2)} \quad (23)$$

$$\boxed{c_{i,A} = U'_{i,}(0) \begin{pmatrix} \alpha(0) \\ (\alpha^\dagger(0))^\top \end{pmatrix} = \mathbf{U}'_{11}(0)_{i,\alpha(0)} + \mathbf{U}'_{12}(0)_{i,} (\alpha^\dagger(0))^\top} \quad (24)$$

$$\begin{aligned} \begin{pmatrix} c_A \\ c_B \end{pmatrix} &= \begin{pmatrix} \frac{1}{2}I & \frac{i}{2}I \\ \frac{1}{2}I & -\frac{i}{2}I \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{W}(0) & \mathbf{V}^*(0) \\ \mathbf{V}(0) & \mathbf{W}^*(0) \end{pmatrix} \begin{pmatrix} \alpha(0) \\ (\alpha^\dagger(0))^\top \end{pmatrix} = \begin{pmatrix} I & I \\ -iI & iI \end{pmatrix} \begin{pmatrix} \mathbf{W}(0) & \mathbf{V}^*(0) \\ \mathbf{V}(0) & \mathbf{W}^*(0) \end{pmatrix} \begin{pmatrix} \alpha(0) \\ (\alpha^\dagger(0))^\top \end{pmatrix} \\ &\equiv U'(0) \begin{pmatrix} \alpha(0) \\ (\alpha^\dagger(0))^\top \end{pmatrix} = \begin{pmatrix} W(0) + V(0) & V^*(0) + W^*(0) \\ i(-W(0) + V(0)) & i(-V^*(0) + W^*(0)) \end{pmatrix} \begin{pmatrix} \alpha(0) \\ (\alpha^\dagger(0))^\top \end{pmatrix} \\ &\equiv \begin{pmatrix} \mathbf{U}'_{11}(0) & \mathbf{U}'_{12}(0) \\ \mathbf{U}'_{21}(0) & \mathbf{U}'_{22}(0) \end{pmatrix} \begin{pmatrix} \alpha(0) \\ (\alpha^\dagger(0))^\top \end{pmatrix} \end{aligned} \quad (25)$$

$$\begin{aligned} &\langle c_{i,A} \alpha^\dagger(2) \rangle \\ &= \left\langle \left\{ \mathbf{U}'_{11}(0)_{i,\alpha(0)} + \mathbf{U}'_{12}(0)_{i,} (\alpha^\dagger(0))^\top \right\} \left\{ \alpha^\dagger(0) \widetilde{W}(2) + \alpha^\top(0) \widetilde{V}(2) \right\} \right\rangle \\ &= (\mathbf{U}'_{12}(0)_{i,} \quad \mathbf{U}'_{11}(0)_{i,}) \left\langle \begin{pmatrix} (\alpha^\dagger(0))^\top \\ \alpha(0) \end{pmatrix} (\alpha^\dagger(0) \quad \alpha^\top(0)) \right\rangle \begin{pmatrix} \widetilde{W}(2) \\ \widetilde{V}(2) \end{pmatrix} \\ &= (\mathbf{U}'_{12}(0)_{i,} \quad \mathbf{U}'_{11}(0)_{i,}) \left(-M^{-1} + \begin{pmatrix} 0 & I \\ 0 & 0 \end{pmatrix} \right) \begin{pmatrix} \widetilde{W}(2) \\ \widetilde{V}(2) \end{pmatrix} \end{aligned} \quad (26)$$

$$M = \begin{pmatrix} \widetilde{F}(2) & -I \\ I & 0 \end{pmatrix} \quad (27)$$

$$M^{-1} = \begin{pmatrix} 0 & I \\ -I & \widetilde{F}(2) \end{pmatrix} \quad (28)$$

$$\mathbf{r} \in B, \mathbf{r} - \boldsymbol{\delta}_\mu \in A,$$

$$\begin{aligned}
\langle \Omega | \sigma_{\mathbf{r}}^{\mu} \beta_{\mathbf{k},1}^{\dagger} | \Omega \rangle &= \left\langle \Omega \left| \sigma_{\mathbf{r}}^{\mu} \left(\frac{1}{\sqrt{N}} \sum_{\mathbf{r}' \in A, \nu} e^{i\mathbf{k} \cdot \mathbf{r}'} d_{\mathbf{r}', \nu}^{\dagger} \phi_{\mathbf{k},1, \nu} \right) \right| \Omega \right\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}', \nu} e^{i\mathbf{k} \cdot \mathbf{r}'} \langle \Omega | \sigma_{\mathbf{r}}^{\mu} d_{\mathbf{r}', \nu}^{\dagger} | \Omega \rangle \phi_{\mathbf{k},1, \nu} \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}', \nu} e^{i\mathbf{k} \cdot \mathbf{r}'} \langle 0_{\chi}, 0_{\alpha(0)} | \left(i b_{\mathbf{r}-\delta_{\mu}+\delta_{\mu}}^{\mu} c_{\mathbf{r}} \right) \alpha^{\dagger}(\mathbf{r}', \nu) \chi_{\mathbf{r}', \nu}^{\dagger} | 0_{\chi}, 0_{\alpha(\mathbf{r}', \nu)} \rangle \phi_{\mathbf{k},1, \nu} \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}', \nu} e^{i\mathbf{k} \cdot \mathbf{r}'} \left\langle 0_{\chi}, 0_{\alpha(0)} \left| i \cdot \frac{1}{i} \left(\chi_{\mathbf{r}-\delta_{\mu}, \mu} - \chi_{\mathbf{r}-\delta_{\mu}, \mu}^{\dagger} \right) c_{\mathbf{r}} \alpha^{\dagger}(\mathbf{r}', \nu) \chi_{\mathbf{r}', \nu}^{\dagger} \right| 0_{\chi}, 0_{\alpha(\mathbf{r}', \nu)} \right\rangle \phi_{\mathbf{k},1, \nu} \quad (29) \\
&= \frac{1}{\sqrt{N}} e^{i\mathbf{k} \cdot (\mathbf{r}-\delta_{\mu})} \left\langle 0_{\chi}, 0_{\alpha(0)} \left| \chi_{\mathbf{r}-\delta_{\mu}, \mu} c_{\mathbf{r}} \alpha^{\dagger}(\mathbf{r}-\delta_{\mu}, \mu) \chi_{\mathbf{r}-\delta_{\mu}, \mu}^{\dagger} \right| 0_{\chi}, 0_{\alpha(\mathbf{r}-\delta_{\mu}, \mu)} \right\rangle \phi_{\mathbf{k},1, \mu} \\
&= \frac{1}{\sqrt{N}} e^{i\mathbf{k} \cdot (\mathbf{r}-\delta_{\mu})} \langle 0_{\chi}, 0_{\alpha(0)} | c_{\mathbf{r}} \alpha^{\dagger}(\mathbf{r}-\delta_{\mu}, \mu) | 0_{\chi}, 0_{\alpha(\mathbf{r}-\delta_{\mu}, \mu)} \rangle \phi_{\mathbf{k},1, \mu} \\
&\equiv \frac{1}{\sqrt{N}} e^{i\mathbf{k} \cdot (\mathbf{r}-\delta_{\mu})} \langle \Psi(0) | \Psi(2) \rangle \langle c_j \alpha^{\dagger}(\mathbf{r}-\delta_{\mu}, \mu) \rangle \phi_{\mathbf{k},1, \mu}
\end{aligned}$$

$$\bullet \quad \circ \quad \langle c_{j,B} \alpha^{\dagger}(2) \rangle$$

$$\boxed{\alpha^{\dagger}(2) = \alpha^{\dagger}(0) \widetilde{W}(2) + \alpha^{\top}(0) \widetilde{V}(2)} \quad (30)$$

$$\boxed{c_{j,B} = U'_{j+N, (0)} \begin{pmatrix} \alpha(0) \\ (\alpha^{\dagger}(0))^{\top} \end{pmatrix} = \mathbf{U}'_{21}(0)_{j, \alpha(0)} + \mathbf{U}'_{22}(0)_{j, (\alpha^{\dagger}(0))^{\top}} \quad (31)$$

$$\begin{aligned}
&\langle c_{j,B} \alpha^{\dagger}(2) \rangle \\
&= \left\langle \left\{ \mathbf{U}'_{21}(0)_{j, \alpha(0)} + \mathbf{U}'_{22}(0)_{j, (\alpha^{\dagger}(0))^{\top}} \right\} \left\{ \alpha^{\dagger}(0) \widetilde{W}(2) + \alpha^{\top}(0) \widetilde{V}(2) \right\} \right\rangle \\
&= (\mathbf{U}'_{22}(0)_{j, \quad \mathbf{U}'_{21}(0)_{j,}} \left\langle \begin{pmatrix} (\alpha^{\dagger}(0))^{\top} \\ \alpha(0) \end{pmatrix} \begin{pmatrix} \alpha^{\dagger}(0) & \alpha^{\top}(0) \end{pmatrix} \right\rangle \begin{pmatrix} \widetilde{W}(2) \\ \widetilde{V}(2) \end{pmatrix} \right) \quad (32) \\
&= (\mathbf{U}'_{22}(0)_{j, \quad \mathbf{U}'_{21}(0)_{j,}} \left(-M^{-1} + \begin{pmatrix} 0 & I \\ 0 & 0 \end{pmatrix} \right) \begin{pmatrix} \widetilde{W}(2) \\ \widetilde{V}(2) \end{pmatrix}
\end{aligned}$$

$$\mathbf{e}_1 = \frac{1}{\sqrt{6}} (1, 1, -2), \quad \mathbf{e}_2 = \frac{1}{\sqrt{2}} (-1, 1, 0) \quad (33)$$

对于 $K = +1$, 已知 $\langle \sigma_i \rangle = (\langle \sigma_i^x \rangle, \langle \sigma_i^y \rangle, \langle \sigma_i^z \rangle)$, $i \in A$ 且与坐标无关,

$$\langle \sigma_i \rangle \cdot \mathbf{e}_1 = \frac{1}{\sqrt{6}} (\langle \sigma_i^x \rangle + \langle \sigma_i^y \rangle - 2 \langle \sigma_i^z \rangle) \quad (34)$$

$$\langle \sigma_i \rangle \cdot \mathbf{e}_2 = \frac{1}{\sqrt{2}} (-\langle \sigma_i^x \rangle + \langle \sigma_i^y \rangle) \quad (35)$$